

Hybrid Preference Scoring

1. Overview

This document has a mathematical formula for Preference Scoring. It requires specific data points from the Supply Side (User Data) and the Demand Side (Job/Market Data).

2. Scoring Formula

The formula aggregates two scores:

Part A: DCE Attribute Utility (S_{attrs})

2.1 Systematic Utility (V) - The Match Calculation

$$V = [\sum(w_i * V_j)] / \sum w_i$$

- Supply Input (w_i): The user's revealed importance.
- Demand Input (V_j): The job's normalized level for that attribute.

2.2 Confidence Scaling (f)

The Supply Side provides a confidence score of completed vignettes. This validates the strength of the user's preference signal before applying it to the Demand Side data.

$$S_{\text{attrs}} = V * f$$

Part B: Work Activity Utility (S_{wa})

This layer maps the User's Likings (Supply) against the Job's Requirements (Demand) using for the work activities

$$S_{\text{wa}} = (1 / N) * \sum [BWS(c) * (I_c / 5) * (L_c / 7)]$$

- **N**: The total count of O*NET activities mapped to the job
- Supply Input [$BWS(c)$]: How much the user likes/dislikes activity c .
- Demand Input [I_c, L_c]: How important activity c is to the job and at what complexity level.

2.3. Final Utility Integration ($\hat{u}$)

$$\text{raw} = S_{\text{attrs}} + S_{\text{wa}}$$

Since both S_{attrs} and S_{wa} are ≤ 1.0 , the maximum raw score is 2.0.

Because the input range is now predictable (0 to 2), we use a Sigmoid Factor of 2.646 (calculations given below).

$$\hat{u} = 1 / (1 + \exp(-(\text{raw} * 2.646)))$$

3. Data Requirements (Supply vs. Demand)

The following inputs are mandatory for the successful execution of the scoring algorithm:

Component	Supply Side (User/Talent)	Demand Side (Job/Employer)
Structural Attributes (Part A)	Preference Vector (w_i): Weights for the 7 attributes derived from DCE trade-offs.	Attribute Levels (V_j): Categorical IDs (e.g., "earn_50k") mapped to 0.0-1.0 scale.
Confidence Metric	Vignette Count: Total number of choice scenarios completed to determine f .	N/A (Standardized across job corpus).
Work Activities (Part B)	BWS Scores: Raw preference scores for specific O*NET activity codes.	O*NET Mapping: Importance (I) and Level (L) scores for the specific role.

Sigmoid Factor calculation

In the Double Normalized implementation, the inputs from both Layer A and Layer B are capped at 1.0. This creates a fixed and predictable range for the total raw score (x):

- Layer A (Attributes) Max: 1.0
- Layer B (Tasks) Max: 1.0
- Theoretical Maximum Raw (x): 2.0

To provide a compelling user experience, a "perfect" technical match ($x = 2.0$) should be represented in the UI as a near-perfect percentage. We have set our anchor point to **99.5%**.

$$\text{Target: } 99.5\% = 0.995$$

We use the Sigmoid to solve for the unknown factor **k** that satisfies our anchor point at $x = 2.0$:

General Formula:

$$\text{Target \%} = 1 / (1 + e^{-(x * k)})$$

Substitution:

$$0.995 = 1 / (1 + e^{-(2.0 * k)})$$

Algebraic Steps:

- $1 + e^{-2.0k} = 1 / 0.995$
- $1 + e^{-2.0k} \approx 1.005025$
- $e^{-2.0k} \approx 0.005025$
- $-2.0k = \ln(0.005025) \approx -5.293$
- **$k = -5.293 / -2.0 \approx 2.646$**